# October 21



Last time we discussed several tools for evaluating linear regression models that have been fit to data.

The simple linear regression model relies on a few key assumptions, and if those assumptions do not actually hold for the data we're analyzing, then the conclusions we draw can be incorrect.

Violated assumptions → incorrect analysis (possibly)

The first assumption we addressed was linearity - that the true population relationship between x and y is linear.

① Linearity: $y = b_0 + b_1 x + \varepsilon$ population relationship

if violated: everything wrong

If this is not actually true, then all of our analysis is incorrect. The least squares estimators for slope and intercept are no longer valid, predictions can be very inaccurate, and inference is incorrect.

The other assumptions related to the error terms in the population model.

error terms $\varepsilon$

② const variance $\sigma^2$

③ normally distributed

④ indep. of each other

} if violated:
* $\hat{\beta}_0, \hat{\beta}_1$ still valid
* model can be used for predicting $\hat{y}$
* inference not valid

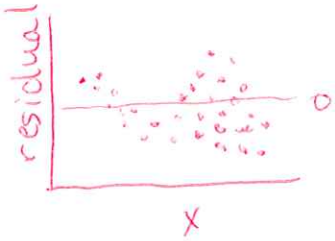
When the assumptions about the error terms are violated, the least squares estimators of the slope and intercept are still valid (and still unbiased), and the model can still be used to predict $\hat{y}_i$ for a given $x_i$. BUT inference related to the model will no longer be correct (hypothesis tests, confidence intervals, prediction intervals).

Checking assumptions: plots of residuals

To evaluate whether any of these assumptions are violated, we look at plots of the residuals from the fitted model.

* scatter plot of $(x, \hat{e})$
  - non-linearity
  - non-const error variance

A scatter plot of pairs $(x_i, \hat{e}_i)$ can reveal issues with nonlinearity, or non-constant variance of the error terms $\varepsilon$.




* histogram or Q-Q plot
  - normality

Non-normality can be checked with a histogram of the residuals, or a "Q-Q" plot, which compares the percentiles of the set of residuals to the percentiles of a normal distribution.

* scatterplot of resids vs index OR lagged resids
  - correlation of error terms

To check for correlation of the error terms, we can try looking at a plot of $(i, \hat{e}_i)$, the residuals vs the index of the data point. OR a plot of the lagged residuals, $(\hat{e}_i, \hat{e}_{i+1})$.

For the 1st two assumptions, and occasionally the 3rd one, using a non-linear transformation of x, or y, or both, may fix the problem.

non-linear transform of x and/or y may resolve ①, ② and maybe ③

By "fix the problem", we mean that if we fit a new linear regression model to the transformed data, the model assumptions WILL be satisfied.

fit linear regression to transformed data

When transformations resolve the assumption violations, they are a helpful strategy, because they let us use all the linear regression modeling tools we already know, just on a modified version of the original data set.

Today we'll spend more time discussing how to use transformations in linear regression modeling. There were a few practical points we mentioned last time:

- popular transformations:
  - natural log, of x and/or y
  - $x^2$, $\sqrt{x}$, $\sqrt{y}$

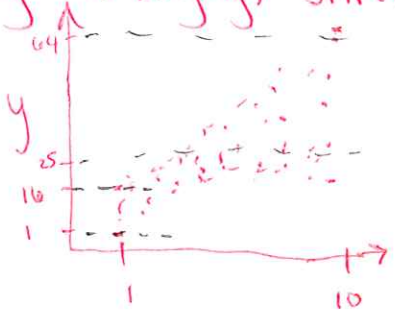
("log" will always refer to natural log aka ln(x) in this class)

- for non-const. error variance, transforming ONLY x doesn't help

If we have a non-constant error variance problem, such that the y values are more spread out for certain x values, transforming only x will not fix the problem.

With a transformation like $\sqrt{y}$ or $\log(y)$, the big values of y get shrunk the most.

$\sqrt{y}$ & $\log(y)$ shrink large values MORE than small ones.




$x=1$ : $y \in [1,16]$ $\sqrt{y} \in [1,4]$

$x=10$ : $y \in [25,64]$ $\sqrt{y} \in [5,8]$

If y is in the range 1 to 16 for small values of x, $\sqrt{y}$ ranges from 1 to 4.

If y is in the range 25 to 64 for large values of x, $\sqrt{y}$ ranges from 5 to 8. The values of $\sqrt{y}$ are spread out the same amt for both x values. Replacing y with $\log(y)$ has a similar effect.

In general, we try a few different transformations, fit a new model, and check the assumptions again.

Suppose we use a log transformation for the y variable. We start with (x,y) data, then replace each $y_i$ value with $\log(y_i)$. Then, we fit a linear model to x and $\log(y)$.

| x | y |
|---|---|
| $x_1$ | $y_1$ |
| $x_2$ | $y_2$ |
| ⋮ | ⋮ |

→

| x | log y |
|---|---|
| $x_1$ | $\log(y_1)$ |
| $x_2$ | $\log(y_2)$ |
| ⋮ | ⋮ |

new model:

$$\log(\hat{y}_i) = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Often, we are interested in the relationship between the original x and y variables, not the transformed version. If we want to predict the y value for a particular x, we can plug that

x value into the fitted model, get $\log(\hat{y})$, then exponentiate to get $\hat{y}$.

x = height (in.)
y = income ($)

$\log(\hat{y}) = 0.06 * \text{height} + 5.91$ → (data circa 2000)

height = 66 → $\log(\hat{y}) = 0.06(66) + 5.91 = 9.87$

For a person w/ height of 66 inches, the model predicts log earnings of 9.87. That isn't very useful to us on the log scale, but exponentiating gives...

$\hat{y} = e^{9.87} \approx 19{,}000$ predicted earnings of $19k.

But what about the slope - what is the interpretation of the 0.06 value now? We think that on average:

$\hat{\beta}_1 = 0.06$ interpretation?

1 inch diff in height, avg 0.06 diff in log income (not very helpful)

people with a one-inch height difference have a 0.06-unit difference in log income. Again, we would prefer to comment on the relationship between height and income, not log income.

Consider $\hat{y}_1$, the prediction for $x = x_1$, and $\hat{y}_2$, the prediction for $x = x_1 + 1$

$(x_1, \hat{y}_1)$ } 1 unit diff in x
$(x_1+1, \hat{y}_2)$ }

$\log(\hat{y}_1) = \hat{\beta}_0 + \hat{\beta}_1 x_1$

$\hat{y}_1 = e^{\hat{\beta}_0 + \hat{\beta}_1 x_1} = e^{\hat{\beta}_0} e^{\hat{\beta}_1 x_1}$

$\hat{y}_2 = e^{\hat{\beta}_0 + \hat{\beta}_1 (x_1+1)} = e^{\hat{\beta}_0} e^{\hat{\beta}_1 x_1 + \hat{\beta}_1} = e^{\hat{\beta}_0} e^{\hat{\beta}_1 x_1} e^{\hat{\beta}_1}$

So the ratio of the predicted y values is...

$$\frac{\hat{y}_2}{\hat{y}_1} = \frac{e^{\hat{\beta}_0} e^{\hat{\beta}_1 x_1} e^{\hat{\beta}_1}}{e^{\hat{\beta}_0} e^{\hat{\beta}_1 x_1}} = e^{\hat{\beta}_1}$$

$$\hat{y}_2 = e^{\hat{\beta}_1} * \hat{y}_1$$

For a one unit (additive) difference in x, we expect y to differ by a multiplicative factor of $e^{\hat{\beta}_1}$.

1 unit x diff ↑

$e^{\hat{\beta}_1}$ multiplicative diff in $\hat{y}$

For people w/ one-inch diff in height, on avg we expect the taller person to have an income $e^{0.06} \approx 1.06$ times larger.

income $e^{0.06} \approx 1.06$ times higher among people 1 inch taller (estimated, on avg)

The hypothesis test for whether $\beta_1 = 0$ is now testing for whether there is a multiplicative difference in y associated with an additive difference in x (NOT whether x and y have a linear relationship).

What if we replace BOTH variables with their natural logs?

| x | y |
|---|---|

→

| log x | log y |
|---|---|
| $\log x_1$ | $\log y_1$ |
| $\log x_2$ | $\log y_2$ |
| ⋮ | ⋮ |

Now the fitted model is:

$$\log \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 \log x_i$$

Suppose our predictor is a tree's diameter in inches, and the response is the total volume of the tree's wood (timber companies want to know this).

x = tree diameter (in.)

y = wood volume ($ft^3$)

$\log \hat{y} = -2.8 + 2.5 \log x$

To predict volume of a tree w/ diam = 10, plug log(10) into the fitted model.

$x = 10 \rightarrow \log \hat{y} = -2.8 + 2.5 * \log(10) \approx 3$

$\hat{y} \approx e^3 \approx 20$

The predicted volume is 20 cubic ft.

What is the interpretation of the slope in this model?

$\hat{\beta}_1$ meaning?

On avg, we expect trees with a one-unit difference in log(diameter) to have a 2.5-unit difference in log(volume).

1 unit diff in log(diam) assoc. w/ 2.5 unit diff in log(volume) (not so helpful)

Again, we really want to comment on the association between the original x and y variables: diameter and volume.

Consider two trees with diameters $x_1$ and $x_2$, and the respective predictions for their volumes, $\hat{y}_1$ and $\hat{y}_2$.

$\log(\hat{y}_1) = \hat{\beta}_0 + \hat{\beta}_1 \log(x_1)$ $\log(\hat{y}_2) = \hat{\beta}_0 + \hat{\beta}_1 \log(x_2)$

$$\log(\hat{y}_2) - \log(\hat{y}_1) = [\hat{\beta}_0 + \hat{\beta}_1 \log(x_2)] - [\hat{\beta}_0 + \hat{\beta}_1 \log(x_1)]$$

$$= \hat{\beta}_1 \log(x_2) - \hat{\beta}_1 \log(x_1)$$

$$= \hat{\beta}_1 [\log(x_2) - \log(x_1)]$$

We can rewrite both sides of this expression using the rules of logarithms:

$$\log\left(\frac{\hat{y}_2}{\hat{y}_1}\right) = \hat{\beta}_1 \log\left(\frac{x_2}{x_1}\right)$$

Then, exponentiating both sides gives us:

$$\frac{\hat{y}_2}{\hat{y}_1} = e^{\hat{\beta}_1 \log(x_2/x_1)} = \left[e^{\log(x_2/x_1)}\right]^{\hat{\beta}_1} = \left(\frac{x_2}{x_1}\right)^{\hat{\beta}_1}$$

In this model, $\hat{\beta}_1$ relates a multiplicative difference in $x$ to a multiplicative difference in $y$.

For trees whose diameters differ by a factor of 1.01, i.e. one of the trees is 1% larger, the expected volumes differ by a factor of $(1.01)^{\hat{\beta}_1} = 1.01^{2.5} \approx 1.025$. In other words, we expect the larger (diameter) tree to have $\approx 2.5\%$ larger volume.

$\frac{x_2}{x_1} = 1.01$ (tree 2 has 1% larger diam)

$\frac{\hat{y}_2}{\hat{y}_1} = 1.01^{\hat{\beta}_1} = 1.01^{2.5} \approx 1.025$

predicted volume $\approx 2.5\%$ larger

For trees with diameters that differ by a factor of 2, we expect the larger tree to have volume $2^{\hat{\beta}_1}$ times larger, on average.

$\frac{x_2}{x_1} = 2$ diam 2 times larger

$\frac{\hat{y}_2}{\hat{y}_1} = 2^{\hat{\beta}_1} = 2^{2.5} \approx 5.7$

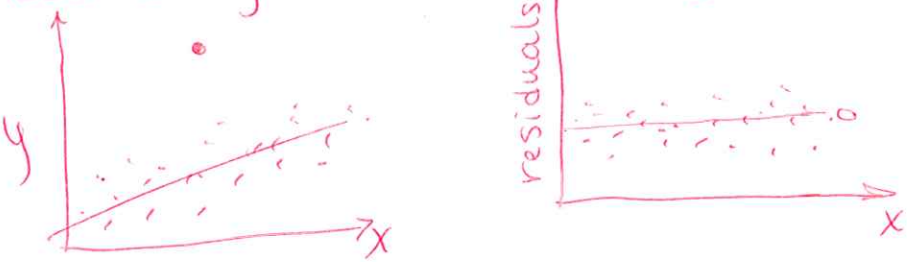
The last two examples were specific to the log transformation. For other nonlinear transformations generally, there isn't a convenient interpretation of the model coefficients in terms of the original (not transformed) x and y variables.

log transformations are special
other transformations don't have nice interpretation for $\hat{b}_1$ (in terms of orig x and y)

Before moving on from model assessment, we need to consider another common problem that arises when fitting linear models to data: outliers.

Broadly, there are two cases to consider: a data point with a y value very far away from the estimated regression line, and a data point with an x value very far away from the rest of the data points.

Case 1: y outlier




If the data point with the unusual y value still has an x value similar to the rest of the data, this does not usually have a large effect on the estimated slope and intercept values.

estimated $\hat{\beta}_0, \hat{\beta}_1$ not strongly affected; $\hat{\sigma}^2$ more affected

It does tend to increase the estimated error term variance ($\hat{\sigma}^2$) a meaningful amount. One data point with a very large residual

can change the standard error of the residuals a lot.

$\hat{\sigma}^2$ used for inference

bigger value → wider confidence intervals

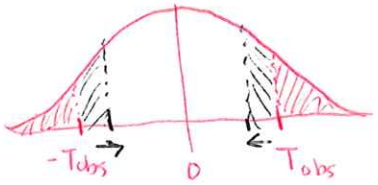
We use the standard error of the residuals to construct confidence intervals and perform hypothesis tests.

$$se(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{(n-1)S_x^2}}$$

A larger value gives us wider confidence intervals, and can change the results of hypothesis tests.

$H_0: \beta_1 = 0$ $H_a: \beta_1 \neq 0$

$$T = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)}$$




For testing whether the true slope = 0, $\hat{\sigma}^2$ appears in the denominator. So a larger value of $\hat{\sigma}^2$ translates to a test statistic T that is closer to zero. This increases the p value for the test, making it harder to reject the null hypothesis (harder to detect whether the slope is truly nonzero)

larger p value

These consequences are undesirable. What can we do about it?

It depends. Sometimes an outlier in the data is a mistake – the value was recorded incorrectly. If the value is impossible based on your knowledge of the variables being studied, you can throw away that data point.

if data impossible/obviously wrong: throw away outlier

Most of the time, we won't know for sure that a data point is actually incorrect, so we can't justify deleting it. Sometimes values DO occur that look very different from what the linear model predicts. If the model can potentially make large mistakes,

usually: don't know if outlier is a mistake, can't delete it

we want to know about it, not pretend it didn't happen.

One thing we often do when we aren't sure, is fit the model to the full data set, then fit another model to the data set with the outlier removed. Comparing the models at least tells us how much of an impact the outlier has on our overall analysis.

another approach: fit 2 models & compare
↳ 1 to full data
↳ 1 to data with outlier removed

how much impact does the outlier have on our analysis?

Case 2: x outlier (aka "leverage point") The other case is an x value far away from the rest of the data. These are usually called "leverage points" instead of outliers. A "leverage point" yanks the estimated regression ~~line~~ line toward itself, like a lever (sort of).

bigger effect on $\hat{\beta}_0$, $\hat{\beta}_1$

The effects of this are less straightforward. If the data point with the extreme ~~x~~ x value happens to be very close to the true population regression line, that helps us estimate the model more accurately. leverage point: $(x^*, y^*)$

- helps estimation if $(x^*, y^*)$ close to true population regression line
- hurts if $(x^*, y^*)$ far from population line

If the ~~y~~ data point with the extreme x value also has a y value far away from the population regression line, that makes the estimated slope and intercept less accurate.

In general, it isn't easy to determine which of these scenarios you're in...

(because we don't know the true population regression line).

in practice: hard to know which scenario you're in

Just like with outlying y values, if the x value for the leverage point is impossible or obviously a mistake, you can remove it.

what to do?
- throw away if impossible/obviously wrong value

But like outlying y values, we often won't know for sure that the leverage point was mistakenly recorded, so we can't justify removing it.

usually: don't know that, can't justify removing

Once again, it may be valuable to fit a linear regression to the full data set, and to the data set with the leverage point removed, and compare the two models to see how much of an impact the leverage point has.

other approach: fit 2 models & compare
↳ 1 to full data
↳ 1 to data w/ leverage point removed.